

Qatar-1b

R-1759

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160419 · created 2026-08-02T20:46:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457497.92623 +/- 0.00079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.05
Transit depth (Rp/Rs)^2	3.4 +/- 1.85 [%]
Orbital Inclination (inc)	80.73 +/- 2.56
Airmass coefficient 1 (a1)	1.004 +/- 0.017
Airmass coefficient 2 (a2)	-0.0013 +/- 0.0074
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	#2 - [396, 216]
Optimal Method	PSF photometry
Transit Duration (day)	0.039 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.05 vs 0.1463 (deviation 0.63σ)
Duration fit vs published	0.039 d vs 0.0692 d (deviation 0.91σ)
Mid-time O–C	2.0 min
Depth SNR	3.1
Residual scatter	1.73 %

Light curve

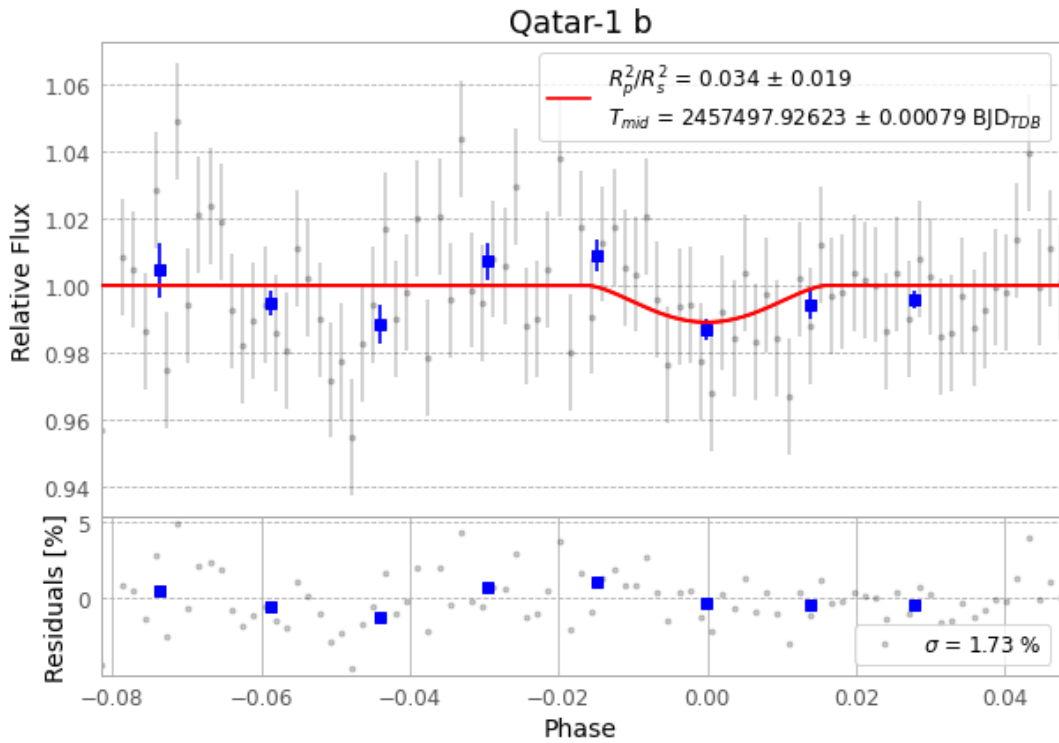


Figure 1 — FinalLightCurve_Qatar-1 b_2016-04-19.png (SHA-256 b0e0e48b7d4836d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 137], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ed8f13b77b336c3f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), Qatar-1160419072412.FITS → Qatar-1160419115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160419.log (652d75333e4e...), FinalLightCurve_Qatar-1 b_2016-04-19.png (b0e0e48b7d48...), FinalLightCurve_Qatar-1 b_2016-04-19.csv (f1c0e135ce88...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:46Z.